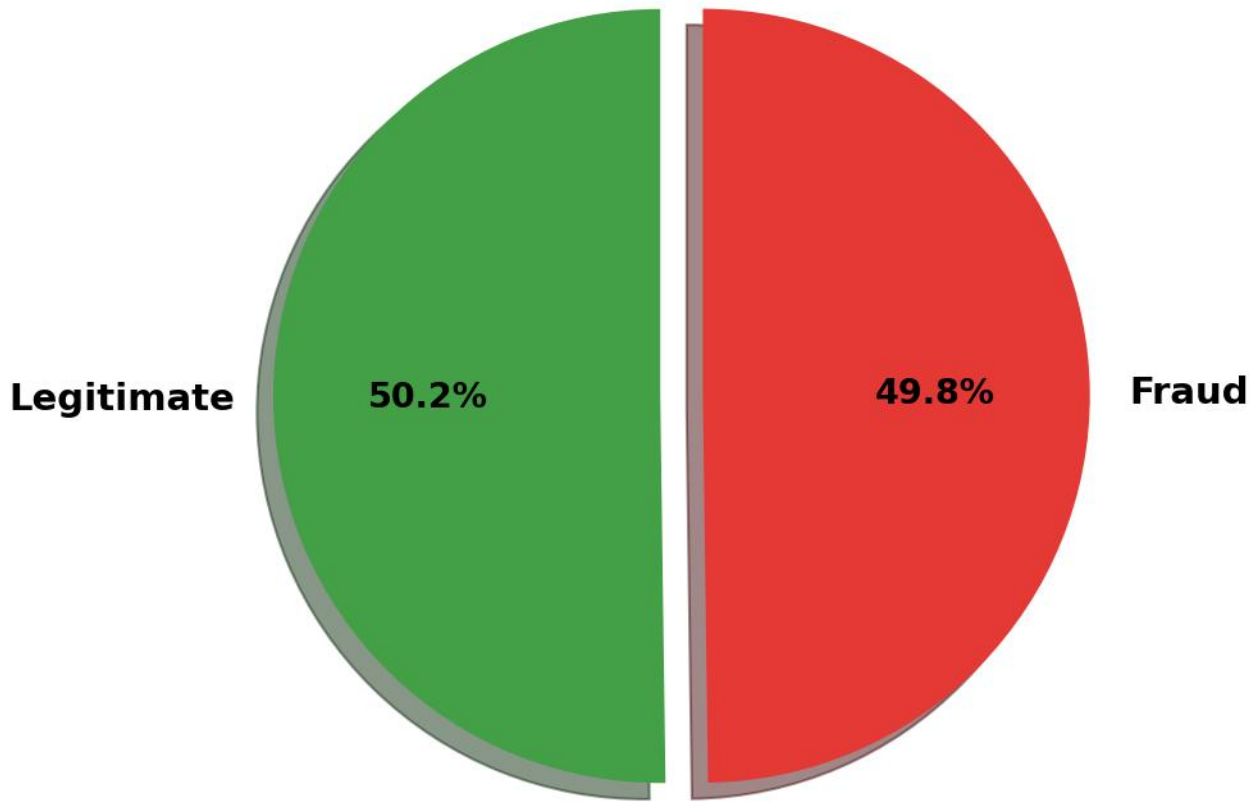


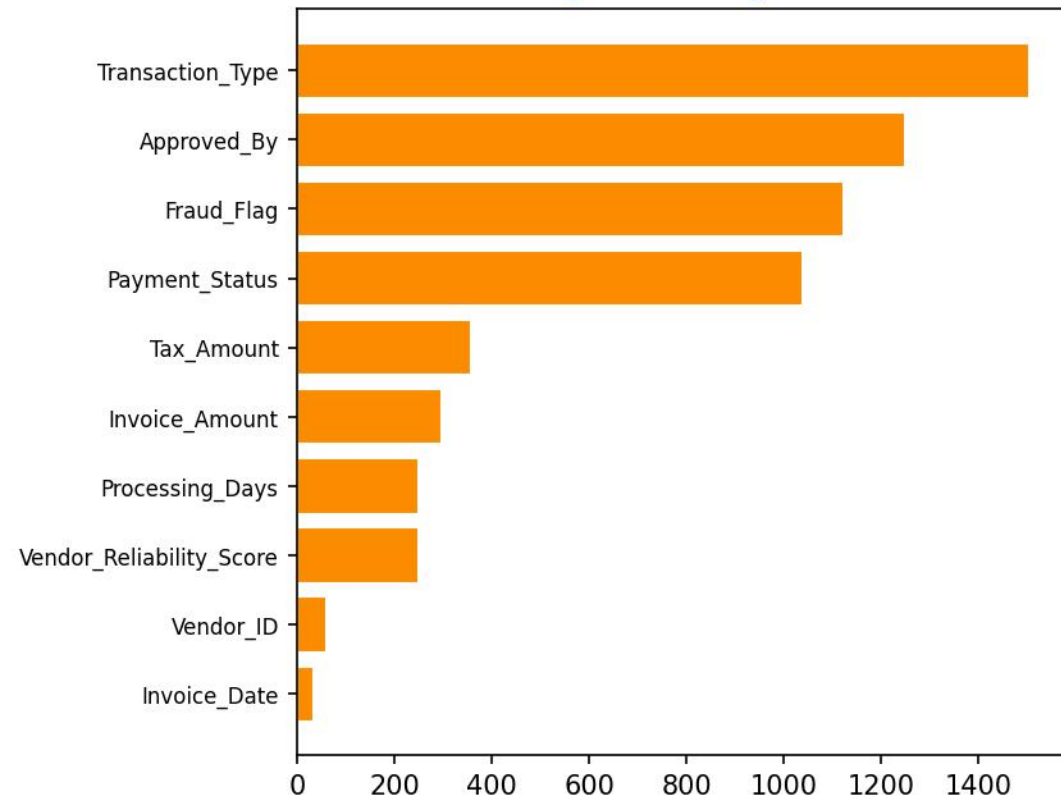
Fraud Distribution in Dataset



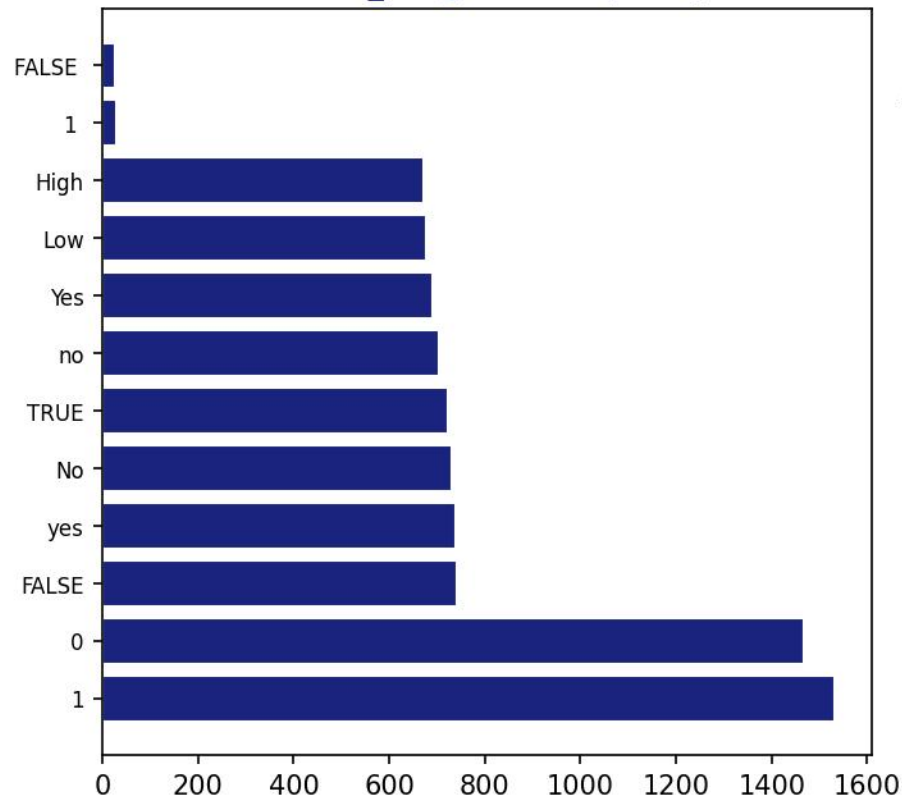
Total: 8,603 records | Fraud: 4,320 | Legitimate: 4,283

Data Quality Assessment

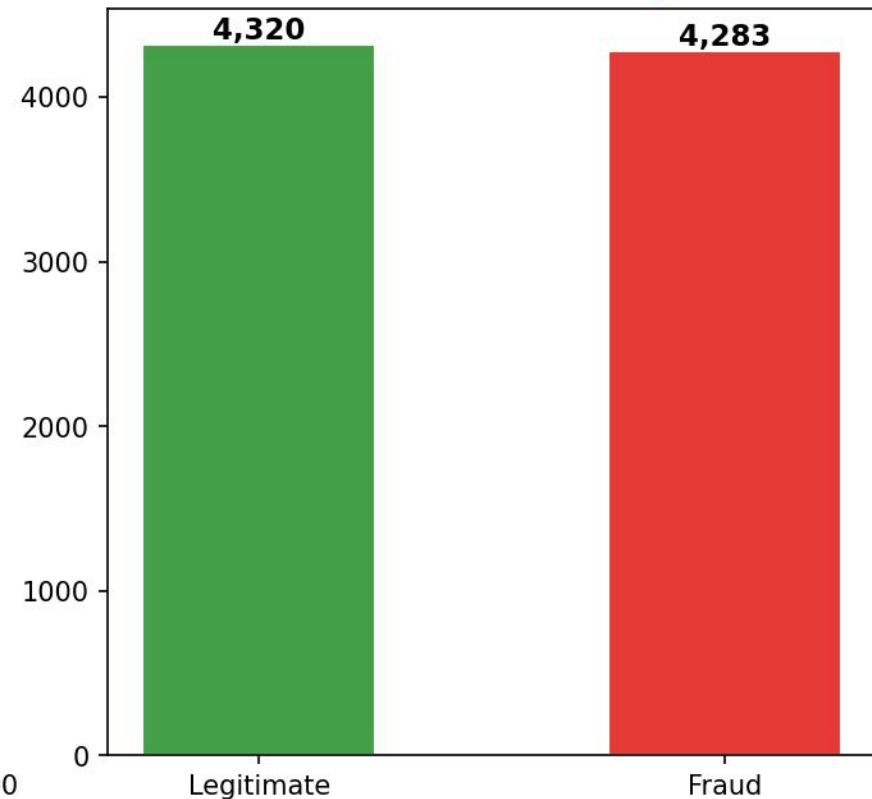
Missing Values by Column



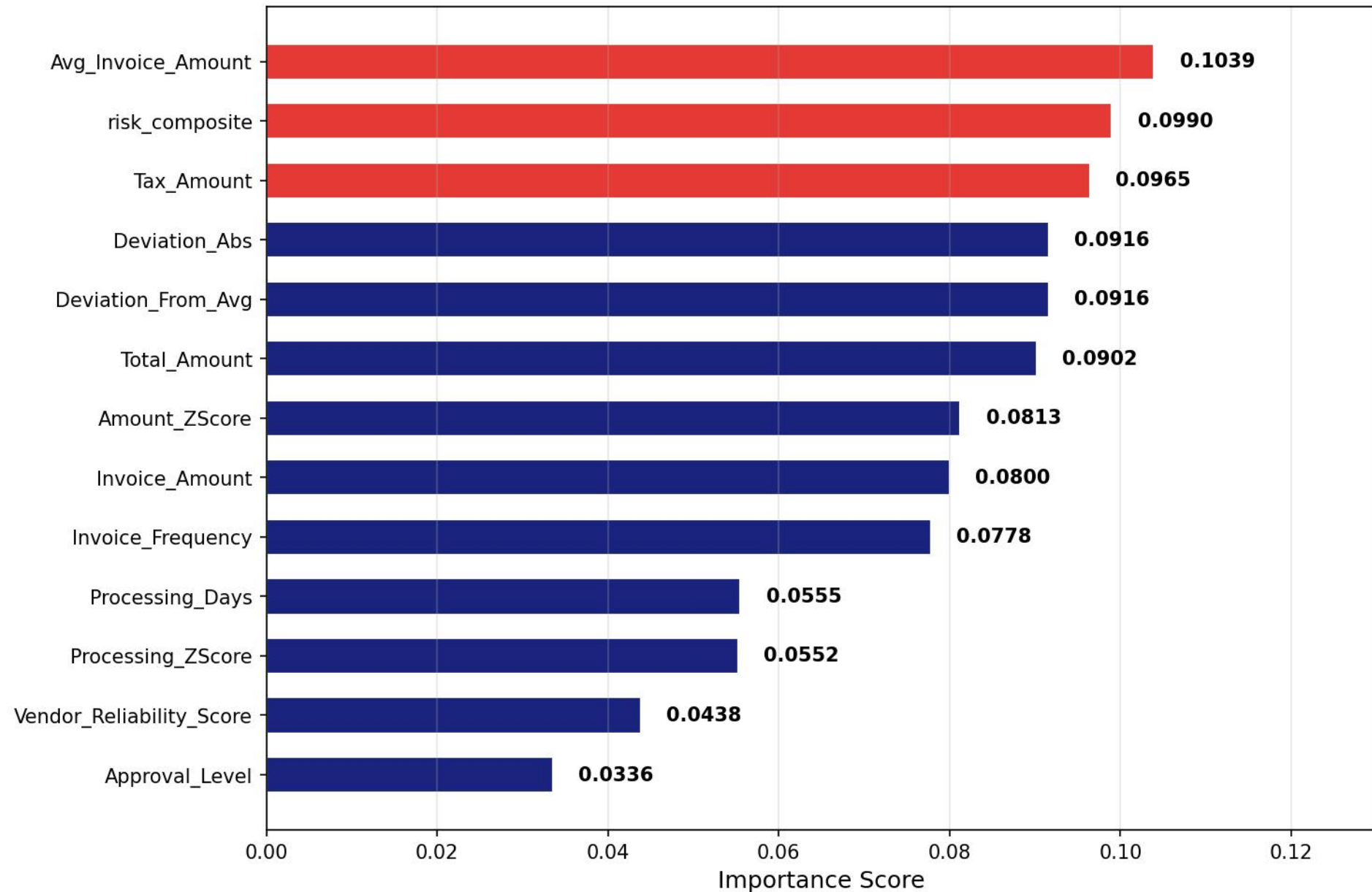
Fraud_Flag Values (Raw)



After Data Cleaning

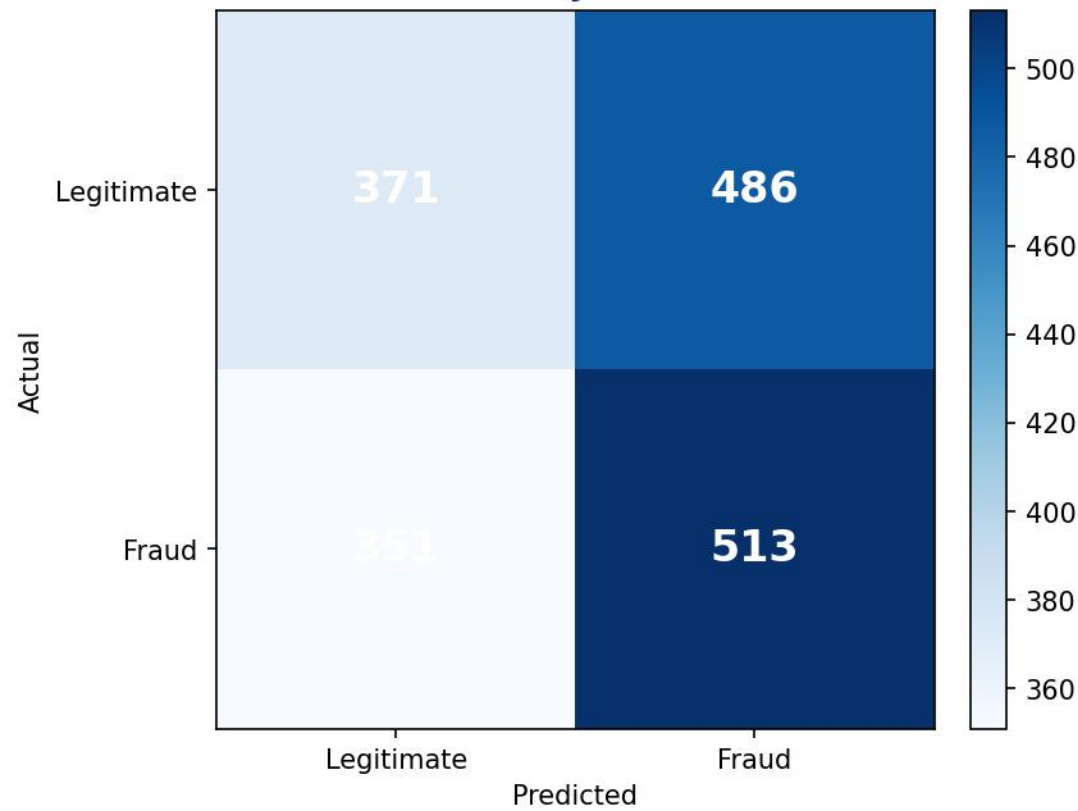


Feature Importance (Random Forest)

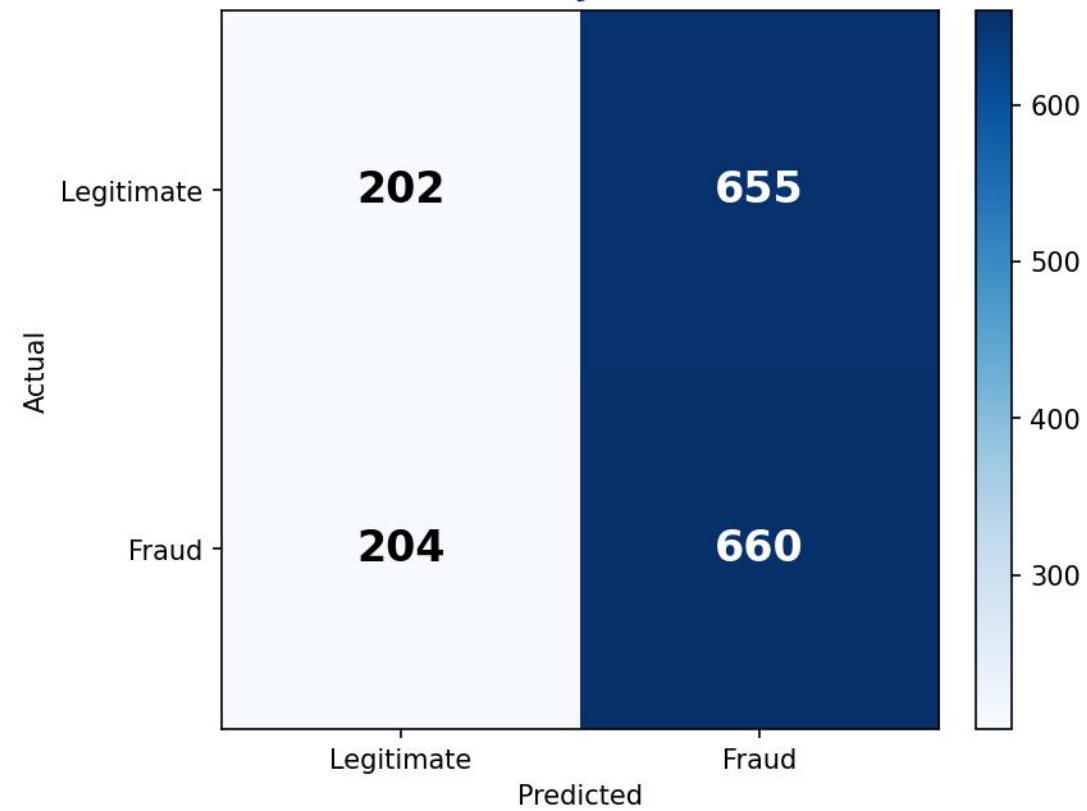


Confusion Matrices- All 4 Models

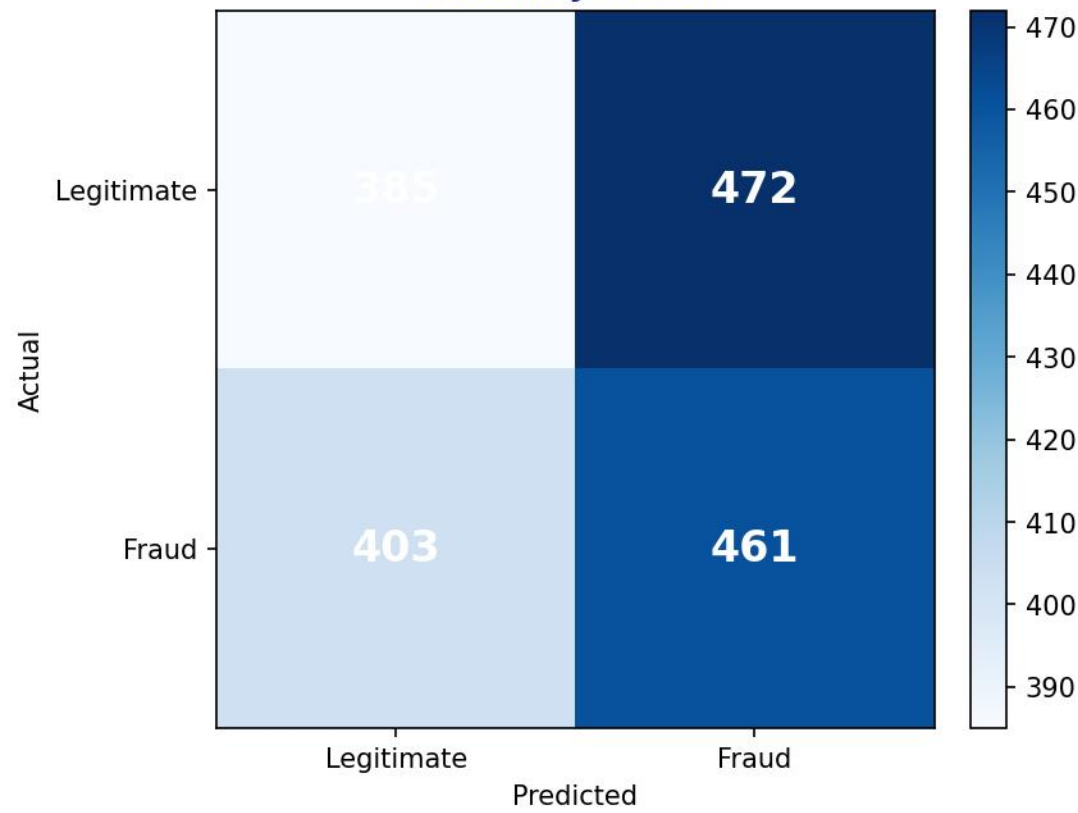
Logistic Regression
Accuracy: 51.4%



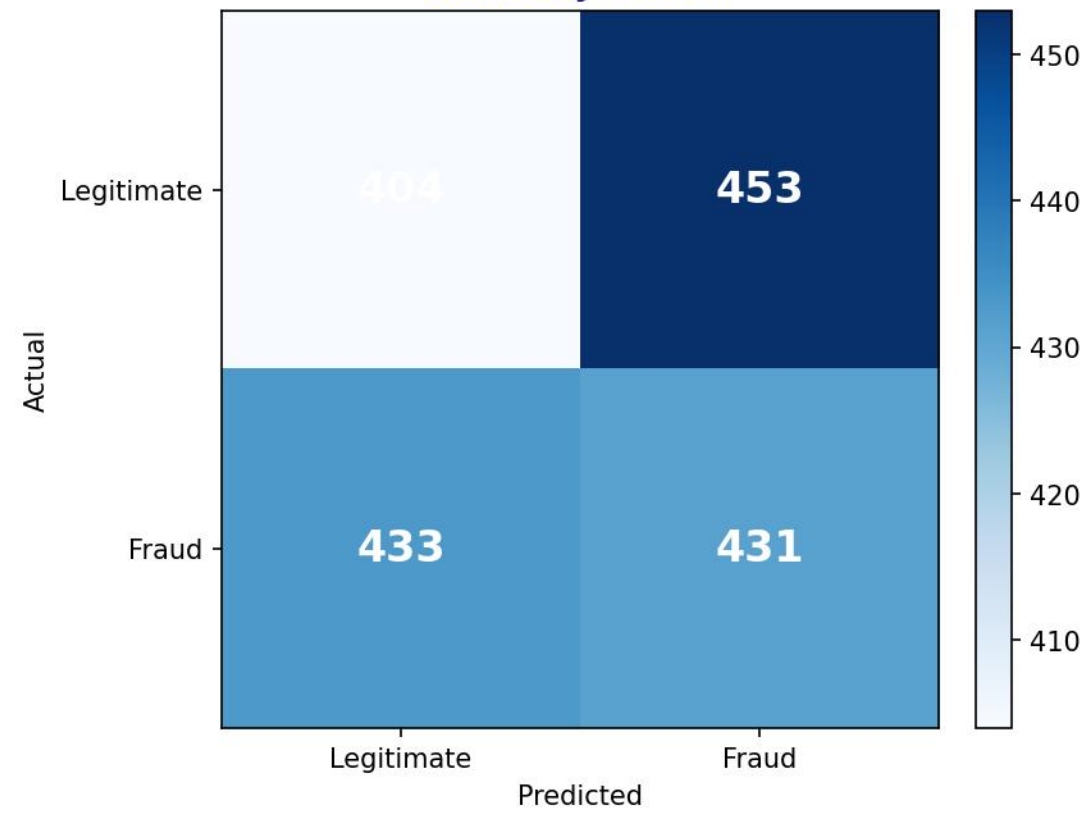
Decision Tree
Accuracy: 50.1%



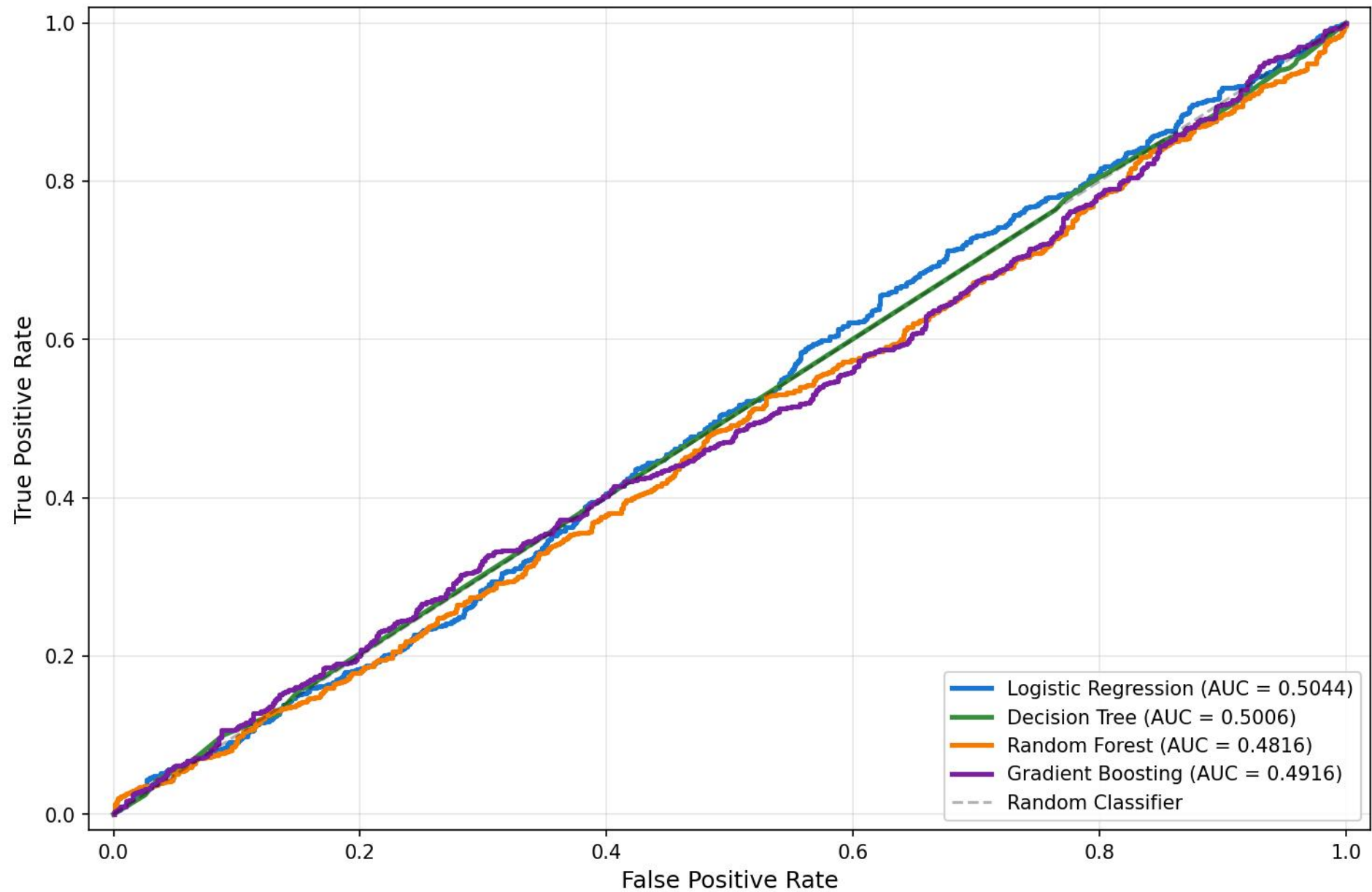
Random Forest
Accuracy: 49.2%



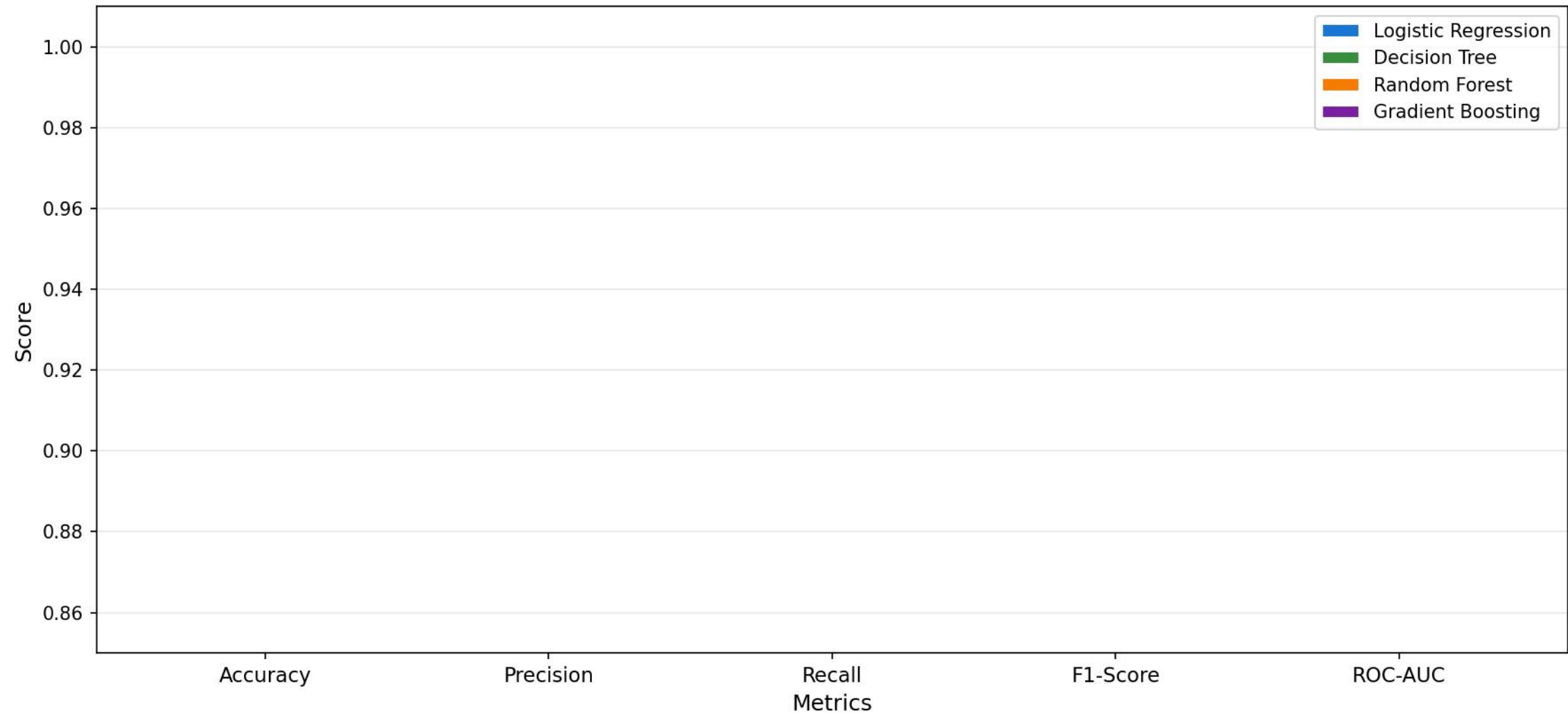
Gradient Boosting
Accuracy: 48.5%



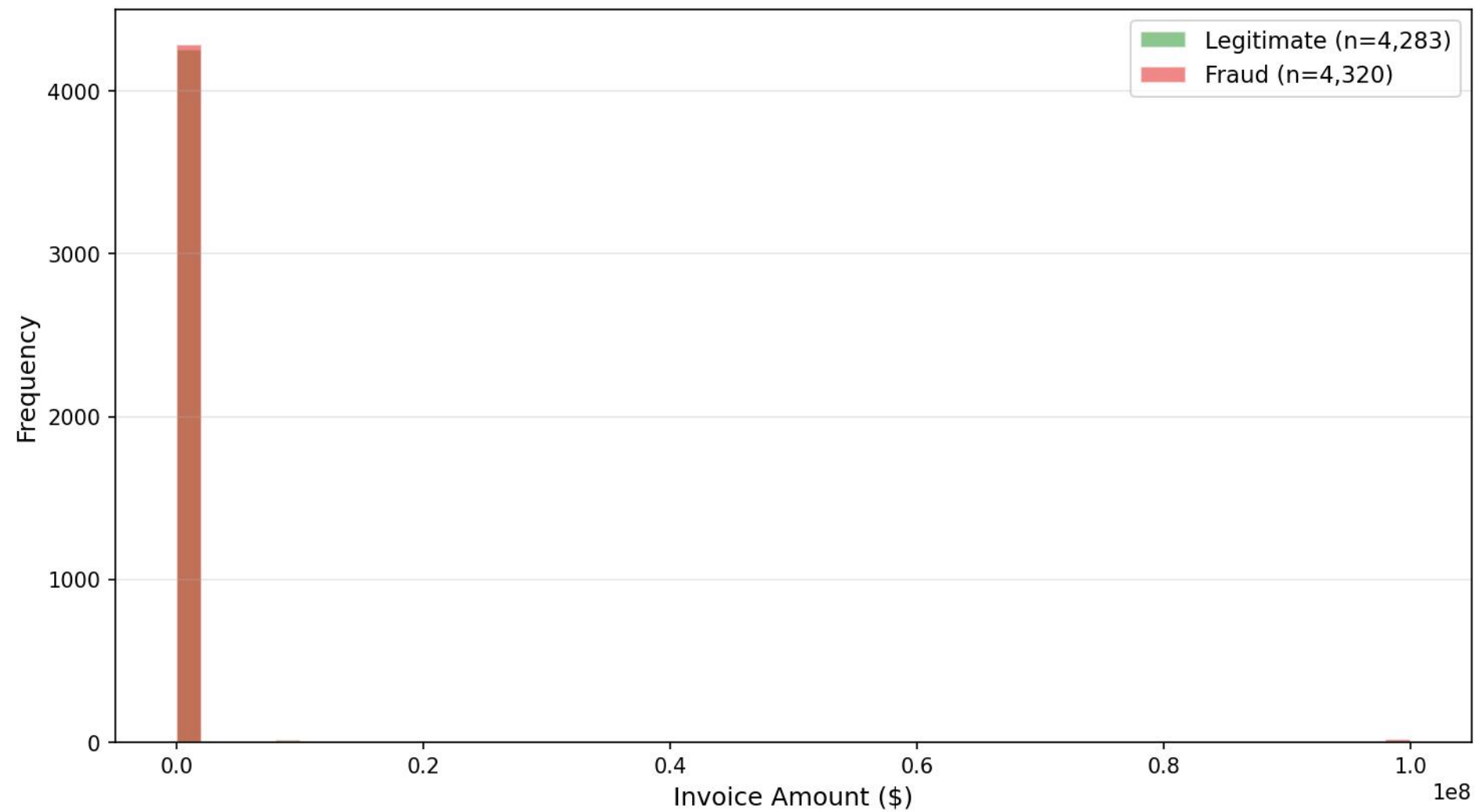
ROC Curves - Model Comparison



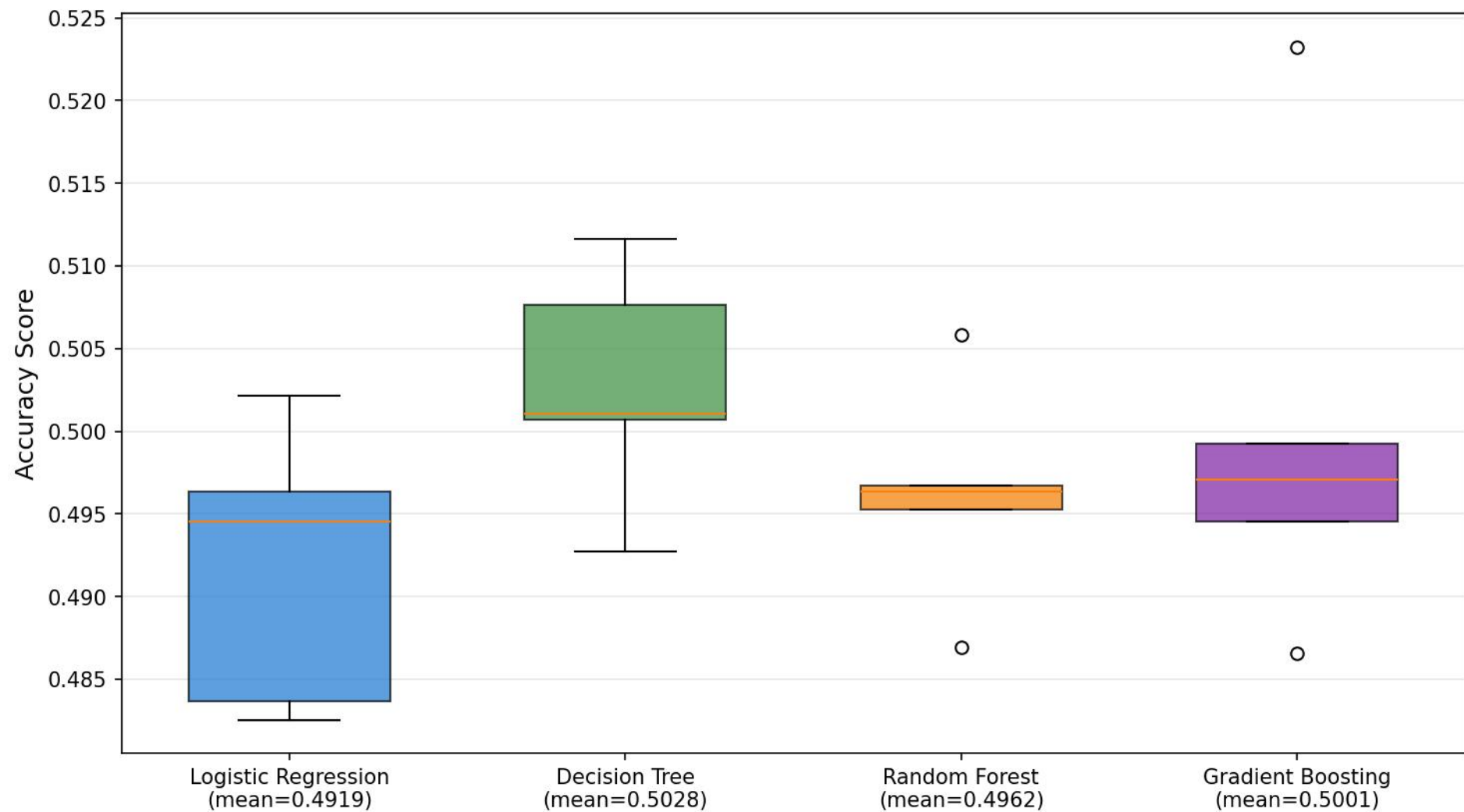
Model Performance Comparison



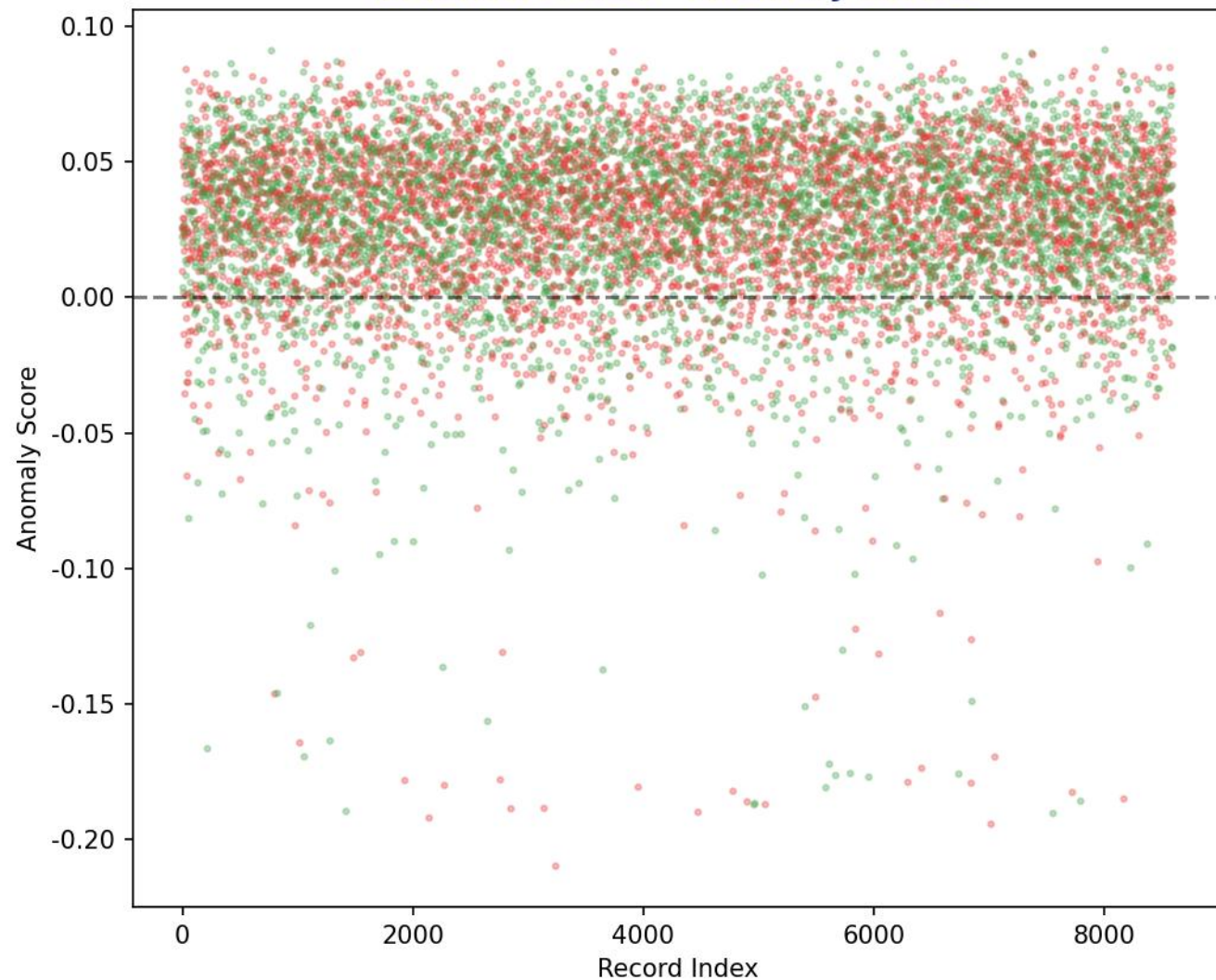
Invoice Amount Distribution: Fraud vs Legitimate



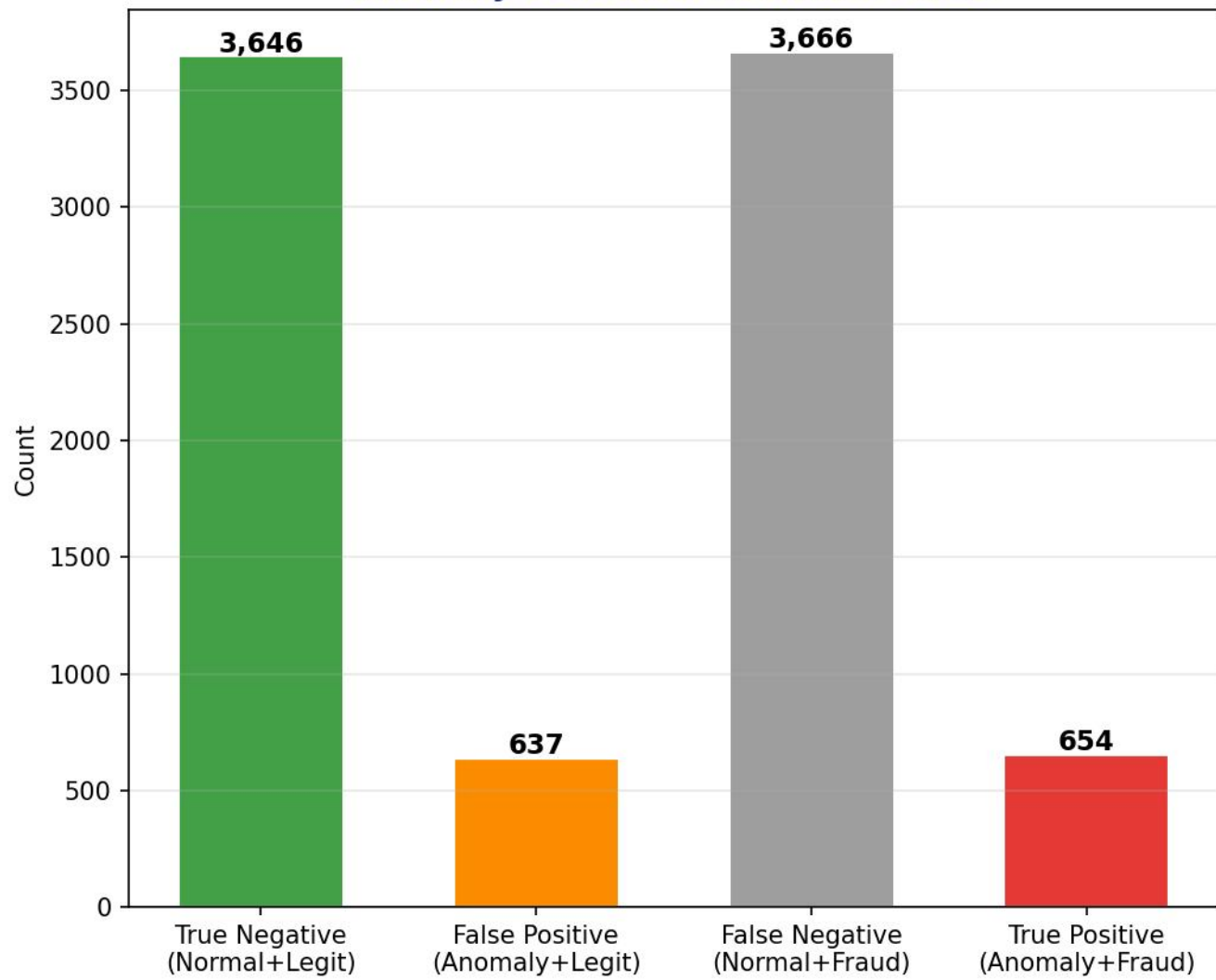
5-Fold Cross-Validation Results



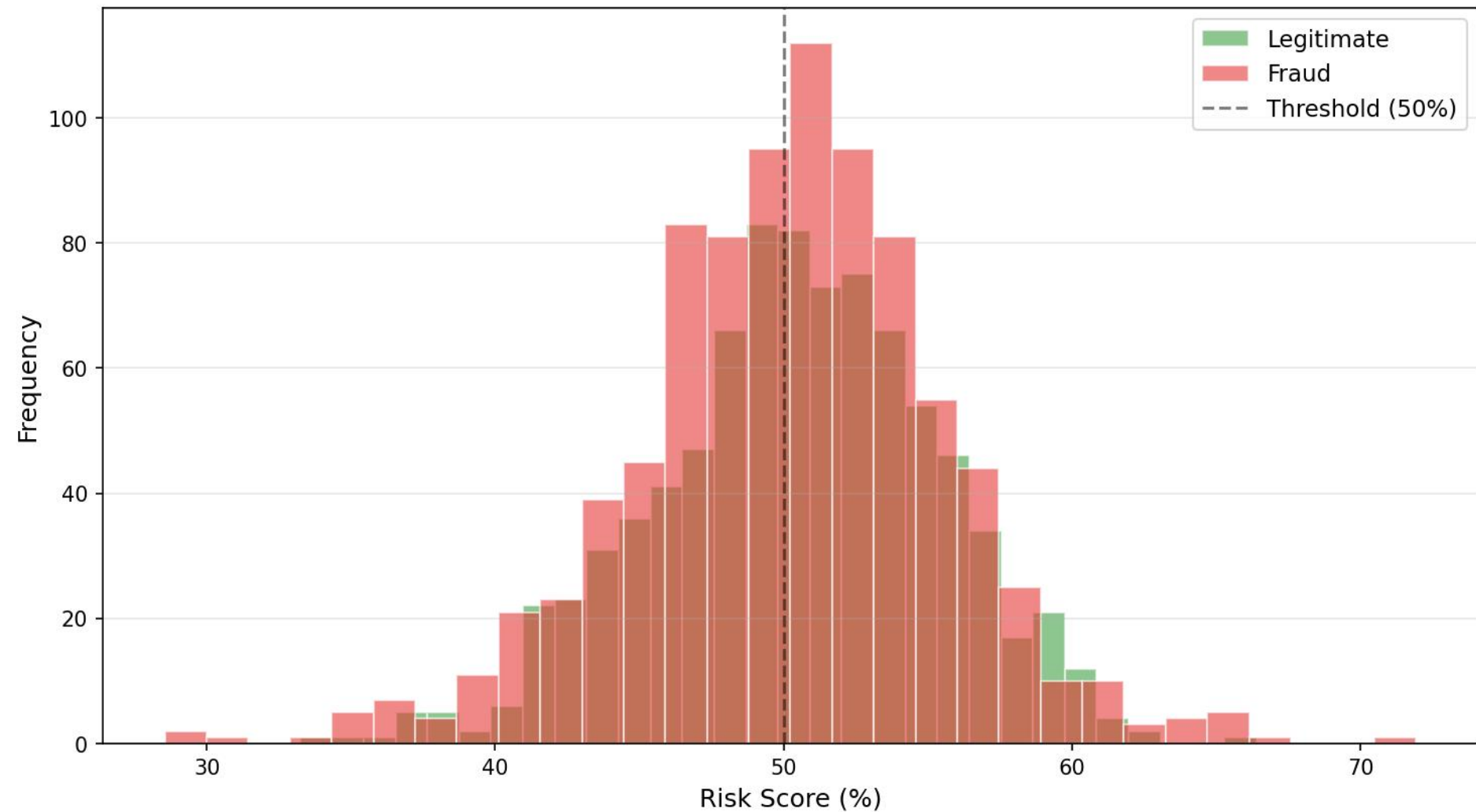
Isolation Forest - Anomaly Scores



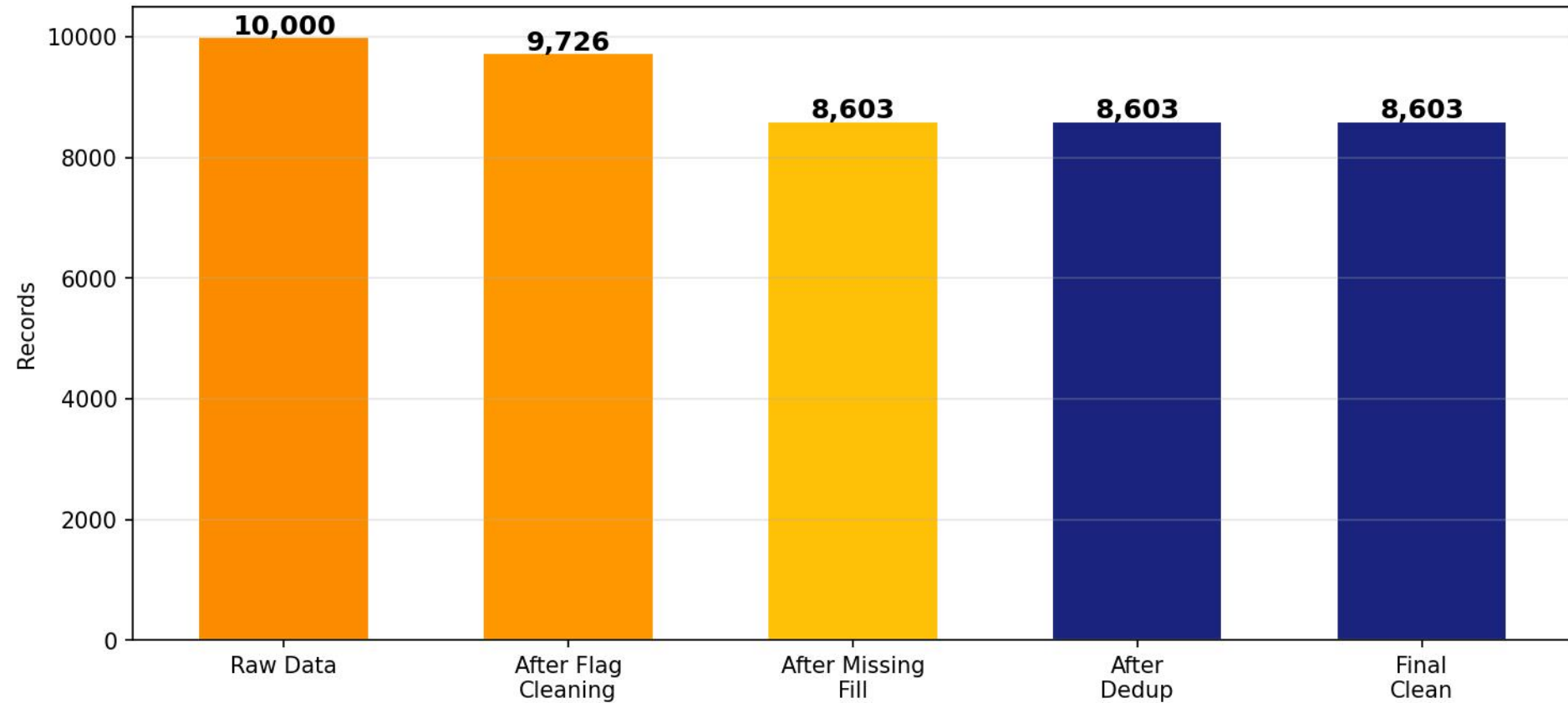
Anomaly Detection vs Actual Fraud



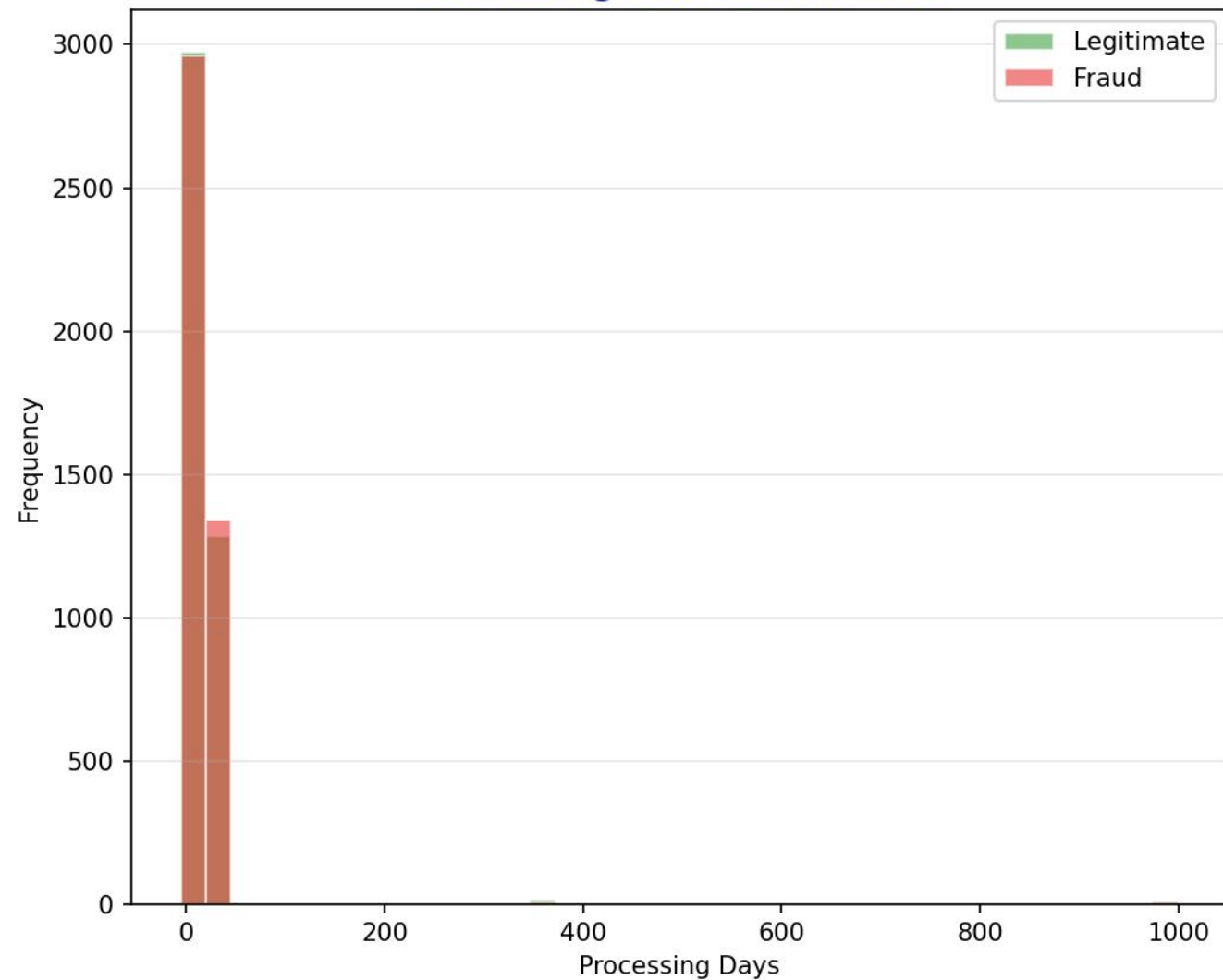
Predicted Risk Score Distribution



Data Cleaning Pipeline



Processing Time Distribution



Fraud by Approval Level (Transaction Logs)

